

acousticRoomResponse

Synthesize room impulse response
Since R2025a

Syntax

```
ir = acousticRoomResponse(room,tx,rx)
ir = acousticRoomResponse(room,tx,rx,Name=Value)
```

Description

`ir = acousticRoomResponse(room,tx,rx)` returns a synthesized impulse response for the given room, transmitter, and receiver. [example](#)

`ir = acousticRoomResponse(room,tx,rx,Name=Value)` specifies options using one or more name-value arguments.

Examples

[collapse all](#)

Synthesize Impulse Response of Shoebox Room

Specify the dimensions of a simple shoebox room. The first dimension is the length along the x-axis, the second dimension is the width along the y-axis, and the third dimension is the height along the z-axis.

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```
roomDimensions = [5, 4, 6];
```

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Specify the cartesian coordinates of the receiver and source within the room.

```
rx = [2, 3.5, 2];
tx = [2, 1.5, 2];
```

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Synthesize the impulse response.

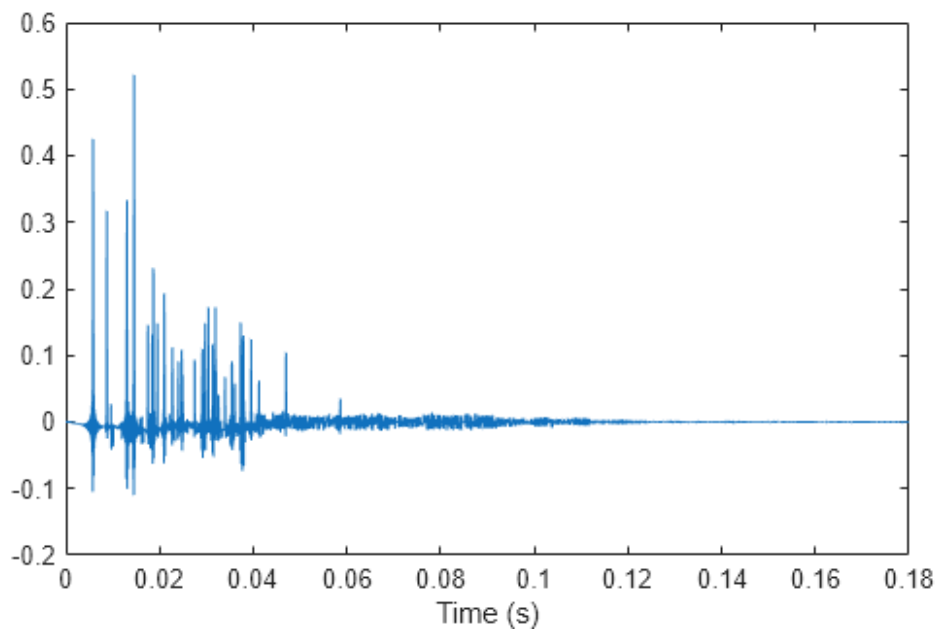
```
ir = acousticRoomResponse(roomDimensions,tx,rx);
```

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Plot the impulse response.

```
fs = 16e3;
t = (0:numel(ir)-1)/fs;
plot(t,ir)
xlabel("Time (s)")
```

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Synthesize Room Impulse Response with Multiple Microphones

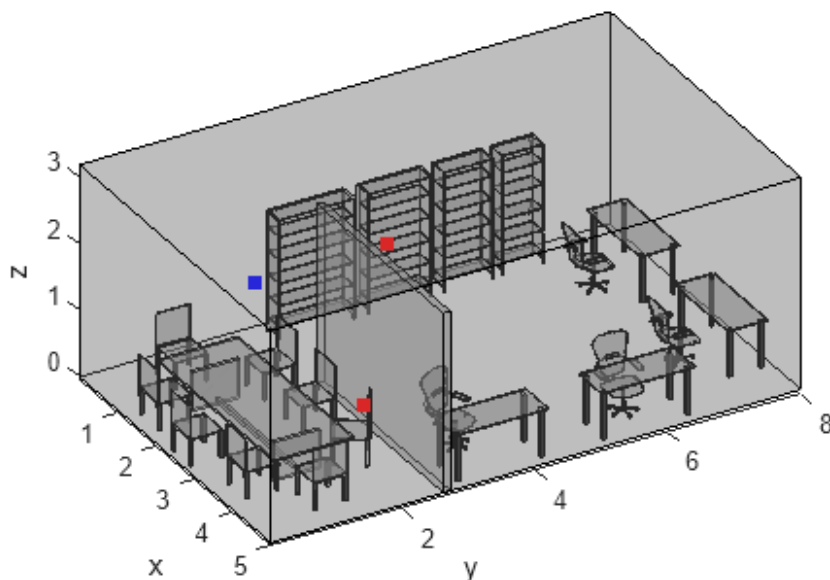
Read in a triangulation object from an STL file using `stlread`. Define the coordinates of the transmitter and two receivers. Use the `viewScene` helper function to visualize the room with the transmitter represented as a blue point and the receivers as red points.

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```
room = stlread("office.stl");
tx = [2, 1.5, 2];
rx = [2, 3.5, 2; 4, 2, 1];
viewScene(room,tx,rx)
```

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Use `acousticRoomResponse` to synthesize the impulse response for both receivers. Set the `ImageSourceOrder` to 2, because the default of 3 is too high for this complex scene.

```
ir = acousticRoomResponse(room,tx,rx,ImageSourceOrder=2);
```

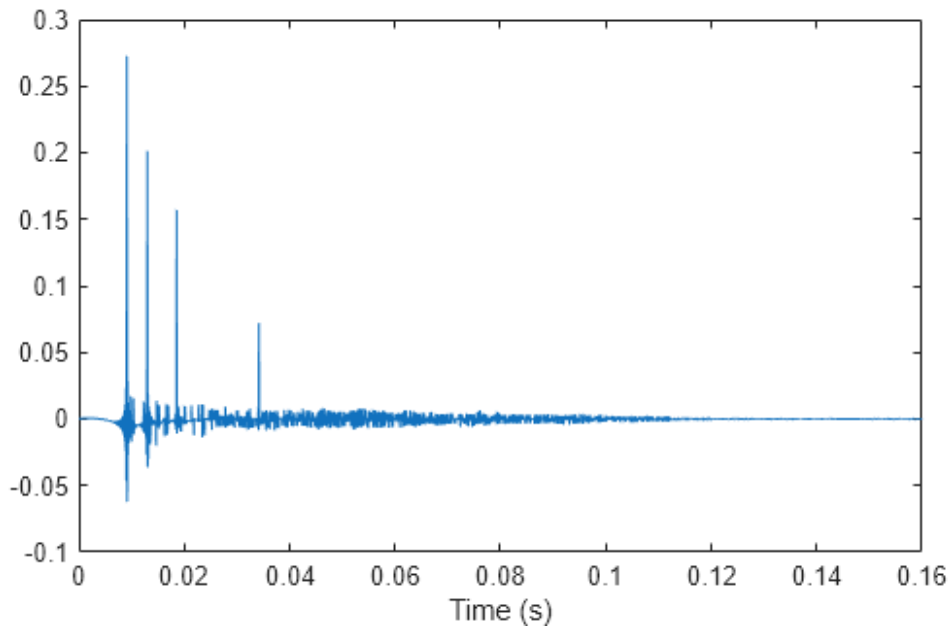
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Filter an audio signal using the first impulse response. Plot the impulse response and listen to the filtered audio. The impulse response for this microphone has no direct line-of-sight to the transmitter, and the sound must reflect around the wall.

```
[x,fs] = audioread("MaleVolumeUp-16-mono-6secs.ogg");
y1 = filter(ir(1,:),1,x);
sound(y1,fs)
```

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```
t = (0:length(ir)-1)/fs;
plot(t,ir(1,:))
xlabel("Time (s)")
```

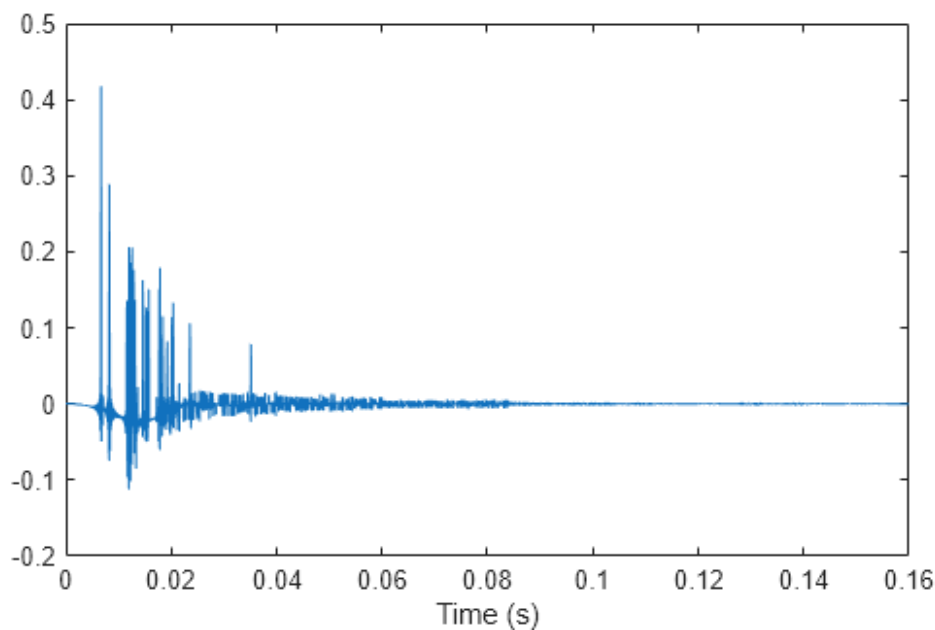


Filter the same signal using the impulse response from the other microphone and listen to the difference. Plot the impulse response. The microphone associated with this impulse response has a direct line-of-sight to the transmitter.

```
y2 = filter(ir(2,:),1,x);
sound(y2,fs)

plot(t,ir(2,:))
xlabel("Time (s)")
```

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Supporting Function

```
function viewScene(room,tx,rx)
% Helper function to visualize scene with source and receiver locations.
trisurf(room,FaceAlpha=0.3,FaceColor=[.5 .5 .5],EdgeColor="none");
view(60, 30);
hold on; axis equal; grid off;
xlabel("x"); ylabel("y"); zlabel("z");
```

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```
% Plot edges
fe = featureEdges(room,pi/20);
numEdges = size(fe, 1);
pts = room.Points;
a = pts(fe(:,1),:);
b = pts(fe(:,2),:);
fePts = cat(1, reshape(a,1,numEdges,3),reshape(b, 1, numEdges, 3), ...
    nan(1,numEdges,3));
fePts = reshape(fePts,[],3);
plot3(fePts(:,1),fePts(:,2),fePts(:,3),"k",LineWidth=.5);
scatter3(tx(1),tx(2),tx(3),"sb","filled");
scatter3(rx(:,1),rx(:,2),rx(:,3),"sr","filled");
hold off
end
```

Specify Room Material Characteristics with `acousticRoomResponse`

Define the dimensions of a shoebox room and the coordinates of the transmitter and receiver. Use `acousticRoomResponse` to synthesize the impulse response with custom absorption and scattering coefficients for the walls of the room. Plot the synthesized response.

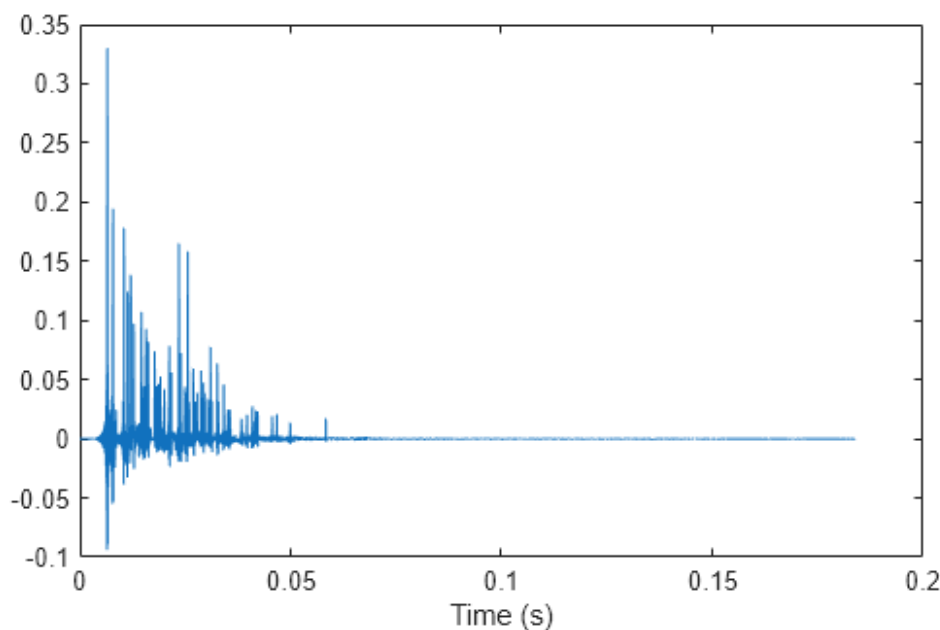
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```
roomDimensions = [6, 5, 2.5];
tx = [3, 3, 1.8];
rx = [5, 4, 1.7];

ir = acousticRoomResponse(roomDimensions,tx,rx,MaterialAbsorption=.5,MaterialScattering=.5);
fs = 16e3;
t = (0:numel(ir)-1)/fs;
plot(t,ir)
xlabel("Time (s)")
```

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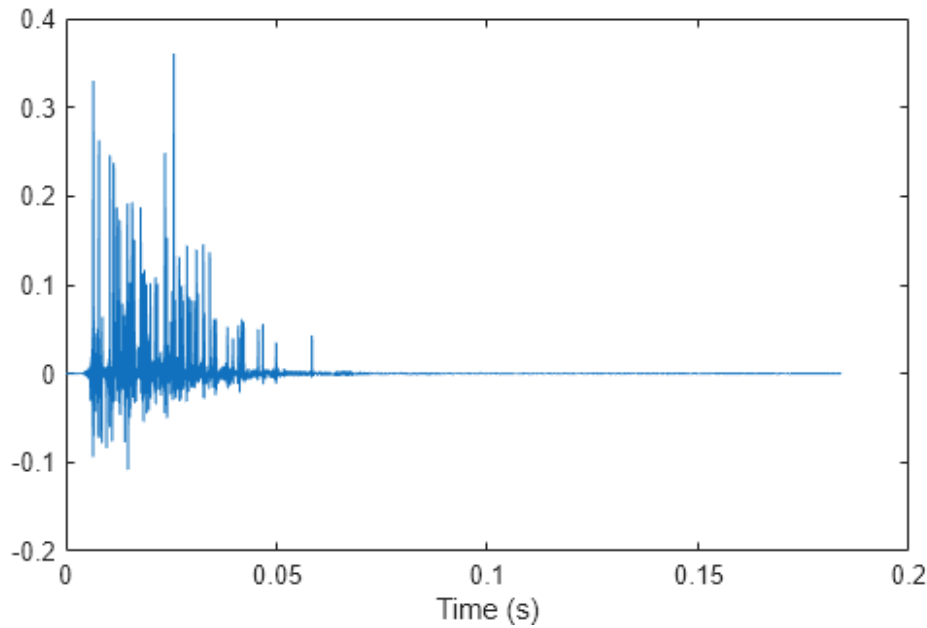


Call `acousticRoomResponse` with the absorption and scattering coefficients set to column vectors to specify the characteristic for each of the surfaces, which are the floor, front, back, left, right, and ceiling of the shoebox room. Plot the resulting impulse response.

```
ir = acousticRoomResponse(roomDimensions,tx,rx,...
    MaterialAbsorption=[.05, .06, .05, .07, .08, .1].',...
    MaterialScattering=[.7, .6, .7, .6, .7, .6].');
plot(t,ir)
```

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```
plot(t,ir)
xlabel("Time (s)")
```

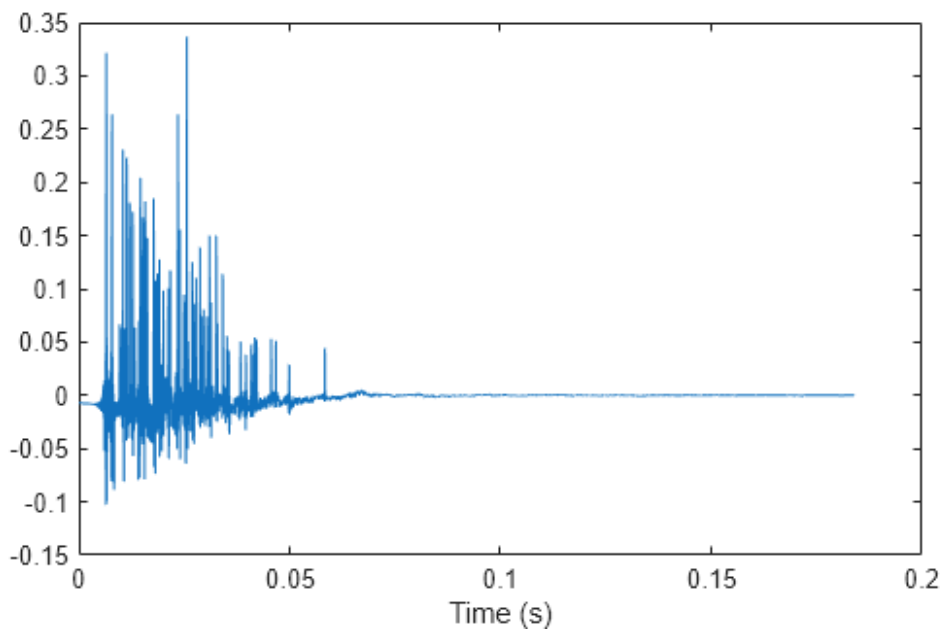


Call `acousticRoomResponse` with the absorption and scattering coefficients set to row vectors to specify the characteristic for each of the frequencies specified by `BandCenterFrequencies`. Plot the resulting impulse response. You can also specify the coefficients as matrices to define the characteristic for each combination of surface and frequency band.

```
ir = acousticRoomResponse(roomDimensions,tx,rx,...
    MaterialAbsorption=[.015, .015, .02, .02, .03, .03, .04, .0
    MaterialScattering=[.2, .2, .3, .5, .6, .6, .7, .7, .7],...
    BandCenterFrequencies=[31.125, 62.5, 125, 250, 500, 1000, 2

plot(t,ir)
xlabel("Time (s)")
```

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Input Arguments

[collapse all](#)

room — Room

1-by-3 vector | triangulation object

The room representation used to synthesize the impulse response, specified as a 1-by-3 vector or a [triangulation](#) object. If you provide a 1-by-3 vector, $[L \ W \ H]$, then the room is a shoebox room with length L along the x-axis, width W along the y-axis, and

height H along the z-axis, where all units are in meters. You can create a `triangulation` object to represent the room using [delaunayTriangulation](#) or you can read in an STL file using [stlread](#).

tx — Transmitter coordinates
1-by-3 vector

Cartesian coordinates of the transmitter in meters, specified as a 1-by-3 vector.
Data Types: `single` | `double`

rx — Receiver coordinates
 N -by-3 vector

Cartesian coordinates of the receiver in meters, specified as an N -by-3 vector, where N is the number of receivers.
Data Types: `single` | `double`

Name-Value Arguments

collapse all

Specify optional pairs of arguments as `Name1=Value1, ..., NameN=ValueN`, where `Name` is the argument name and `Value` is the corresponding value. Name-value arguments must appear after other arguments, but the order of the pairs does not matter.

Example: `ir = acousticRoomResponse(room,tx,rx,ImageSourceOrder=5)`

SampleRate — Sample rate of the impulse response (Hz)
16000 (default) | positive scalar

Sample rate of the impulse response in Hertz, specified as a positive scalar.
Data Types: `single` | `double` | `int8` | `int16` | `int32` | `int64` | `uint8` | `uint16` | `uint32` | `uint64`

SoundSpeed — Speed of sound (m/s)
343 (default) | positive scalar

Speed of sound in meters per second, specified as a positive scalar.
Data Types: `single` | `double` | `int8` | `int16` | `int32` | `int64` | `uint8` | `uint16` | `uint32` | `uint64`

Algorithm — Synthesis algorithm
"hybrid" (default) | "image-source" | "stochastic ray tracing"

Impulse response synthesis algorithm specified as "image-source", "stochastic ray tracing", or "hybrid".

- Set `Algorithm` to "image-source" to use the image-source method for synthesizing the impulse response [1]. For more information about the image-source algorithm, see [Room Impulse Response Simulation with the Image-Source Method and HRTF Interpolation](#).
- Set `Algorithm` to "stochastic ray tracing" to use stochastic ray tracing to synthesize the impulse response [3]. For more information about this algorithm, see [Room Impulse Response Simulation with Stochastic Ray Tracing](#).
- Set `Algorithm` to "hybrid" to combine the image-source and stochastic ray tracing methods to synthesize the impulse response.

Data Types: `char` | `string`

ImageSourceOrder — Image-source maximum order

3 (default) | nonnegative integer

Image-source maximum order, specified as a nonnegative integer. The order specifies the maximum number of reflections a ray can make by bouncing off multiple surfaces before reaching the receiver. A higher image-source order increases the computational complexity of this method.

This argument applies only when [Algorithm](#) is "image-source" or "hybrid".

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

NumStochasticRays — Number of rays for stochastic ray tracing

1280 (default) | nonnegative integer

Number of rays to use for the stochastic ray tracing method, specified as a nonnegative integer. A higher number of rays increases the accuracy of the impulse response but also increases the computational complexity of this method.

This argument applies only when [Algorithm](#) is "stochastic ray tracing" or "hybrid".

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

MaxNumRayReflections — Maximum number of reflections per stochastic ray

10 (default) | nonnegative integer

Maximum number of reflections per stochastic ray, specified as a nonnegative integer. This argument applies only when [Algorithm](#) is "stochastic ray tracing" or "hybrid".

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

ReceiverRadius — Radius of the spherical receiver (m)

0.5 (default) | nonnegative scalar

Radius of the spherical receiver in meters, specified as a nonnegative scalar. This argument applies only when [Algorithm](#) is "stochastic ray tracing" or "hybrid".

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

AirAbsorption — Air absorption coefficient

0 (default) | scalar in range [0,1] | vector

Air absorption coefficient, specified as a scalar in the range [0,1]. You can also specify this argument as a vector to define the air absorption coefficients for each of the [BandCenterFrequencies](#).

Data Types: single | double

MaterialAbsorption — Material absorption coefficients

[.02, .02, .03, .03, .04, .05, .05] (default) | vector | matrix

Material absorption coefficients, specified as a 1-by- N vector, M -by- N matrix, or M -by-1 vector of values in the range [0,1], where N is the number of frequency bands specified by [BandCenterFrequencies](#) and M is the number of surfaces in the [room](#). For triangulation objects, M corresponds to the number of triangles. The absorption coefficient describes the proportion of energy lost during reflection.

Data Types: `single` | `double`

MaterialScattering – Material scattering coefficients

[4e-4, 1e-3, 5e-3, .02, .05, .06, .07] (default) | vector | matrix

Material scattering coefficients, specified as a 1-by- N vector, M -by- N matrix, or M -by-1 vector of values in the range [0,1], where N is the number of frequency bands specified by [BandCenterFrequencies](#) and M is the number of surfaces in the [room](#). For `triangulation` objects, M corresponds to the number of triangles. The scattering coefficient describes the proportion of energy that is scattered as opposed to perfectly reflected when bouncing off a surface.

Data Types: `single` | `double`

BandCenterFrequencies – Center frequencies of bandpass filters (Hz)

[125, 250, 500, 1000, 2000, 4000, 8000] (default) | positive vector

Center frequencies of the bandpass filters in Hertz, specified as a vector of positive values.

Data Types: `single` | `double`

Output Arguments

[collapse all](#)

ir – Impulse response

N -by- L matrix

Impulse response, returned as an N -by- L matrix where N is the number of microphones specified by [rx](#), and L is the length of the impulse response, which depends on the [SampleRate](#), [Algorithm](#), and the characteristics of the room.

References

- [1] Allen, Jont B., and David A. Berkley. "Image Method for Efficiently Simulating Small-Room Acoustics." *The Journal of the Acoustical Society of America* 65, no. 4 (April 1, 1979): 943–50. <https://doi.org/10.1121/1.382599>.
- [2] Peterson, Patrick M. "Simulating the Response of Multiple Microphones to a Single Acoustic Source in a Reverberant Room." *The Journal of the Acoustical Society of America* 80, no. 5 (November 1, 1986): 1527–29. <https://doi.org/10.1121/1.394357>.
- [3] Schröder, Dirk. "Physically Based Real-Time Auralization of Interactive Virtual Environments." PhD Thesis, RWTH Aachen University, 2011.

Version History

Introduced in R2025a

See Also

[stlread](#) | [deLaunayTriangulation](#)

Topics

[Room Impulse Response Simulation with Stochastic Ray Tracing](#)

[Room Impulse Response Simulation with the Image-Source Method and HRTF Interpolation](#)